

MANAGEMENT INFORMATION SYSTEM

Unit-III: Strategic & Project Planning for MIS & Conceptual System Design

Comprehensive Study Notes

PART A: STRATEGIC & PROJECT PLANNING FOR MIS

1. BUSINESS PLANNING

1.1 Introduction to Business Planning

Definition: Business planning is the process of defining an organization's objectives, strategies, and action plans to achieve its goals. It provides a roadmap for the organization's future direction.

Why Business Planning is Important for MIS

Reason	Description
Alignment	Ensures MIS supports business goals
Resource Allocation	Guides investment in IS
Priority Setting	Identifies which systems to build
Risk Management	Identifies and mitigates risks
Performance Measurement	Provides benchmarks for success

1.2 Levels of Business Planning

Level	Description	Timeframe	Responsibility
Strategic	Long-term direction, vision, mission	3-5 years	Top management
Tactical	Departmental plans, resource allocation	1-2 years	Middle management
Operational	Day-to-day operations, short-term goals	Daily/Monthly	Lower management

1.3 Strategic Planning

Definition: Strategic planning is the process of defining the organization's direction and making decisions on allocating resources to pursue this direction.

Key Components

Component	Description
Vision Statement	What the organization wants to become
Mission Statement	Purpose and reason for existence
Goals	Broad, long-term objectives
Strategies	How to achieve goals
Objectives	Specific, measurable targets

Strategic Planning Process

1. Define vision and mission
2. Set long-term goals
3. Conduct environmental analysis (SWOT)
4. Formulate strategies
5. Develop implementation plans
6. Monitor and evaluate

1.4 SWOT Analysis

Definition: SWOT Analysis is a strategic planning tool that evaluates an organization's Strengths, Weaknesses, Opportunities, and Threats.

SWOT Analysis for MIS Planning

	Strengths	Weaknesses
Internal	Skilled IT staff Modern infrastructure Strong data management User-friendly systems	Outdated hardware Legacy systems Security gaps Poor integration
	Opportunities	Threats
External	New technologies (AI, Cloud) Digital transformation Growing data needs Mobile adoption	Cybersecurity threats Competitor systems Rapid technology change Budget constraints

1.5 Business Strategy and MIS

Business Strategy	MIS Strategy	Examples
Cost Leadership	Efficiency, Automation	TPS, ERP, Inventory systems
Differentiation	Innovation, Customer Focus	CRM, Analytics, Personalization
Focus Strategy	Specialized, Niche	Industry-specific solutions

2. MIS RESPONSES

2.1 Introduction to MIS Responses

Definition: MIS Responses refer to how the Management Information System adapts and responds to the changing needs of the organization's business strategy, environment, and user requirements.

2.2 Types of MIS Responses

Type	Description	Example
Strategic Response	Major changes in system direction	New ERP implementation
Tactical Response	Medium-level adjustments	Adding new modules
Operational Response	Day-to-day changes	Report modifications

2.3 MIS Response to Business Changes

Business Change	MIS Response	Example
New Product Launch	Product data setup, Sales tracking, Inventory management	Adding product master, sales reports
Market Expansion	New location setup, Multi-currency support, Language support	Branch system, global reporting
Regulatory Change	Compliance reporting, Data privacy, Audit trails	New reports, security features
Competitive Threat	Competitive analysis, Pricing optimization, Customer analytics	BI tools, pricing systems
Technology Shift	System upgrade, New platforms, Integration	Cloud migration, mobile apps
Customer Needs	CRM enhancements, Personalization, Service automation	Customer portal, chatbots

3. MIS PLANNING - GENERAL & DETAILED

3.1 Introduction to MIS Planning

Definition: MIS Planning is the process of defining the information systems needed to support business strategies and operations, and developing a plan to acquire and implement these systems.

3.2 Need for MIS Planning

Reason	Description
Strategic Alignment	Ensure systems support business goals
Resource Allocation	Efficient use of IT budget
Priority Setting	Identify which projects to start first
Risk Management	Identify and mitigate risks
Integration	Ensure systems work together
Change Management	Prepare for organizational change

3.3 General MIS Planning

Definition: General planning involves the high-level, strategic decisions about the overall direction of MIS in the organization.

General Planning Components

Component	Description	Questions to Ask
Vision	Long-term direction	Where do we want MIS to be in 5 years?
Mission	Purpose of MIS	Why does MIS exist?
Objectives	Specific goals	What should MIS achieve?
Scope	Coverage of systems	What systems are included?
Architecture	Technical framework	What technologies to use?
Governance	Decision-making	Who decides what?
Standards	Guidelines	What standards to follow?

3.4 Detailed MIS Planning

Definition: Detailed planning involves specific, actionable plans for implementing the general MIS strategy.

Detailed Planning Process

Step 1: Define Project Scope

Aspect	Description
What's Included	Systems, modules, features
What's Excluded	What is not part of the project
Deliverables	What will be produced
Success Criteria	How success is measured

Step 2: Resource Allocation

Resource Type	Considerations
People	Number needed, Skills required, Availability
Hardware	Servers, computers, Network equipment, Peripheral devices
Software	Licenses, Development tools, Third-party software
Facilities	Office space, Server room, Training rooms

Step 3: Timeline Development

Activity	Duration	Dependency
Analysis	4 weeks	-
Design	6 weeks	Analysis
Development	12 weeks	Design
Testing	4 weeks	Development
Training	2 weeks	Testing
Implementation	2 weeks	Training

Step 4: Budget Planning

Cost Category	Description
Hardware	Servers, workstations, network
Software	Licenses, development tools
Personnel	Salaries, consultants
Training	User training, technical training
Maintenance	Support, upgrades
Contingency	Unexpected costs (10-20%)

Step 5: Risk Management

Risk Type	Example	Probability	Impact	Mitigation
Technical	System integration failure	Medium	High	Prototype testing
Resource	Key staff leaving	Low	High	Cross-training
Schedule	Project delays	High	Medium	Buffer time
Budget	Cost overruns	Medium	High	Contingency fund

3.5 Comparison: General vs Detailed Planning

Aspect	General Planning	Detailed Planning
Level	Strategic	Tactical
Timeframe	Long-term (3-5 years)	Short-term (1 year)
Responsibility	Top management	Project managers
Granularity	High-level	Detailed
Flexibility	Less detailed	More specific
Focus	What and Why	How and When
Output	Strategic plan	Project plans

PART B: CONCEPTUAL SYSTEM DESIGN

4. INTRODUCTION TO SYSTEM DESIGN

4.1 What is System Design?

Definition: System design is the process of defining the architecture, components, modules, interfaces, and data for a system to satisfy specified requirements.

Two Phases of System Design

Phase	Focus	Output
Conceptual Design	What the system does	Design report, specifications
Detailed Design	How it does it	Detailed specifications, code

Conceptual Design vs Detailed Design

Aspect	Conceptual Design	Detailed Design
Focus	What the system does	How it does it
Level	High-level	Low-level
Output	Design report, specifications	Detailed specifications, code
Users	Management, users	Technical team
Decision	Feasibility, scope	Implementation details

5. CONCEPTUAL SYSTEM DESIGN PROCESS

5.1 Overview of Conceptual Design

Definition: Conceptual system design is the initial phase of system design that defines the overall system architecture, objectives, constraints, and information needs without going into implementation details.

5.2 Steps in Conceptual System Design

1. **Define Problem** - Identify and understand the problem
2. **Set System Objectives** - Define what the system must achieve
3. **Establish System Constraints** - Identify limitations
4. **Determine Information Needs & Sources** - Identify data requirements
5. **Develop Alternative Conceptual Designs** - Generate possible solutions
6. **Prepare the Design Report** - Document the design

6. STEP 1: DEFINE PROBLEM

6.1 Introduction

Definition: The first step in conceptual design is to clearly define the problem that the system is intended to solve.

6.2 Techniques for Problem Definition

Technique	Description	Example
Interviews	Talk to stakeholders	Users, managers, customers
Observation	Watch current processes	Shadowing staff
Document Review	Study existing documents	Reports, forms, manuals
Survey/Questionnaire	Collect quantitative data	User satisfaction
Root Cause Analysis	Find underlying causes	Why-why analysis
Process Mapping	Map current process	Flowcharts

6.3 Problem Statement Format

Component	Description	Example
Problem	What is wrong	"Inventory records are inaccurate"
Impact	What is affected	"Stockouts and overstocking"
Causes	Why it happens	"Manual data entry errors"
Scope	What is included	"All warehouse inventory"
Objectives	What to achieve	"Accurate real-time inventory"

6.4 Example: Problem Definition

Element	Description
Problem	Inaccurate inventory records
Impact	Production delays due to stockouts, Excess inventory costs, Customer dissatisfaction
Causes	Manual data entry errors, No real-time updates, Poor record-keeping
Scope	Raw materials, Work-in-progress, Finished goods
Desired Outcome	95%+ inventory accuracy, Real-time visibility, Automated updates

7. STEP 2: SET SYSTEM OBJECTIVES

7.1 Introduction

Definition: System objectives are the specific, measurable goals that the system must achieve to solve the defined problem and meet organizational needs.

7.2 Types of System Objectives

Type	Description
Functional Objectives	What the system must do (features, functions)
Performance Objectives	How well the system performs (speed, capacity)
Operational Objectives	How the system operates (usability, availability)
Security Objectives	Protection of data and resources
User Objectives	User satisfaction and ease of use

7.3 Characteristics of Good Objectives (SMART)

Criterion	Description	Example
Specific	Clearly defined	"Reduce inventory errors"
Measurable	Quantifiable	"Reduce errors by 90%"
Achievable	Realistic	"With barcode scanning"
Relevant	Important to business	"Save & 5 lakh/year"
Time-bound	Has deadline	"Within 6 months"

7.4 Example: System Objectives for Inventory Management System

Objective Type	Specific Objective
Functional	Track all inventory items, Generate reorder alerts, Produce stock reports
Performance	Process transactions in < 2 seconds, Handle 10,000+ inventory items, 99.5% uptime
Operational	Barcode scanning for data entry, User-friendly interface, Integration with accounting
Security	Role-based access control, Audit trails, Data encryption
User	Reduce data entry time by 50%, Improve user satisfaction to 90%

8. STEP 3: ESTABLISH SYSTEM CONSTRAINTS

8.1 Introduction

Definition: System constraints are the limitations and restrictions that affect the design and implementation of the system.

8.2 Types of Constraints

A. Technical Constraints

Constraint	Description	Example
Hardware	Available equipment	Existing servers must be used
Software	Must use specific platforms	Must run on Windows Server
Network	Bandwidth limitations	Limited internet connectivity
Data	Data format compatibility	Must integrate with legacy database
Integration	Must work with existing systems	Must connect to SAP

B. Economic Constraints

Constraint	Description	Example
Budget	Maximum spend	Total budget: ₹ 50 lakh
Cost	Operational costs	Ongoing costs limited to ₹ 2 lakh/year
ROI	Return on investment	Must pay back in 3 years

C. Operational Constraints

Constraint	Description	Example
Staff Skills	Available expertise	Only 2 IT staff available
Process	Existing procedures	Must work with current workflow
Environment	Working conditions	Must work in remote locations

D. Schedule Constraints

Constraint	Description	Example
Deadline	Mandatory completion date	Go-live by April 1
Milestones	Phased delivery	First phase in 3 months

E. Legal/Regulatory Constraints

Constraint	Description	Example
Data Privacy	Data protection laws	Must comply with IT Act
Industry Regulations	Sector-specific rules	Banking: RBI guidelines
Security	Required security levels	Must meet ISO 27001

F. Organizational Constraints

Constraint	Description	Example
Culture	Organizational values	Must support our culture
Policy	Internal rules	Must follow IT policy
Politics	Power dynamics	Must have department approval

9. STEP 4: DETERMINE INFORMATION NEEDS & SOURCES

9.1 Introduction

Definition: This step involves identifying what information the system must provide, to whom, in what format, and where the data will come from.

9.2 Types of Information Needs

Type	Description	Users
Operational Information	Day-to-day transactions	Operational staff
Tactical Information	Departmental performance	Middle management
Strategic Information	Long-term trends and decisions	Top management

9.3 Information Sources

A. Internal Sources

Source	Description	Examples
TPS	Transaction processing systems	Sales, purchases, payments
Operational Systems	Day-to-day systems	Inventory, payroll, HR
Databases	Stored data	Customer, product, employee
Documents	Paper records	Contracts, forms, reports

B. External Sources

Source	Description	Examples
Market Research	Industry data	Reports, surveys
Government	Public data	Statistics, regulations
Suppliers	Partner data	Catalogs, price lists
Customers	User feedback	Reviews, surveys
Competitors	Competitive intelligence	Websites, reports
Internet	Web data	News, social media

9.4 Example: Information Needs Analysis for Inventory Management

User Level	Information Need	Source	Format	Frequency
Store Staff	Current stock levels	Database	On-screen display	Real-time
Store Manager	Inventory reports	Database	Printed report	Daily
Purchasing Manager	Reorder alerts	System	Email/Alert	As needed
Finance Manager	Inventory valuation	Database	Monthly report	Monthly
CEO	Inventory trends	Data analysis	Dashboard	Weekly

10. STEP 5: DEVELOP ALTERNATIVE CONCEPTUAL DESIGNS & DOCUMENTATION

10.1 Introduction

Definition: This step involves generating multiple possible conceptual design solutions for the system and documenting each alternative for evaluation.

10.2 Types of Alternative Solutions

Alternative	Description	When to Use
In-house Development	Build system internally	Unique requirements, core competency
Off-the-Shelf Software	Buy existing software	Standard requirements, limited budget
Outsourced Development	Hire external team	Limited in-house skills
Cloud/SaaS Solution	Subscription-based	Cost-effective, quick deployment
Open Source	Free software with customization	Technical expertise available
Hybrid Approach	Combination	Complex projects with multiple needs

10.3 Evaluating Alternatives

Criteria	Description	Weight
Cost	Total cost of ownership	25%
Time	Implementation timeline	20%
Functionality	Meets requirements	25%
Flexibility	Easy to modify	10%
Integration	Works with other systems	10%
Risk	Implementation risk	10%

Evaluation Matrix

Criteria	Weight	In-house	Off-shelf	Cloud
Cost	25%	6	8	9
Time	20%	4	9	8
Functionality	25%	9	6	7
Flexibility	10%	9	4	5
Integration	10%	8	6	7
Risk	10%	5	8	7
Weighted Total	100%	7.0	7.0	7.2

Decision: Alternative 3 (Cloud Solution) is slightly better.

11. STEP 6: PREPARE THE DESIGN REPORT

11.1 Introduction

Definition: The Design Report (or Conceptual Design Report) is a comprehensive document that captures all the work done in the conceptual design phase and serves as the basis for detailed design and implementation.

11.2 Purpose of Design Report

Purpose	Description
Communication	Explains system to all stakeholders
Documentation	Records decisions and rationale
Approval	Gets management sign-off
Basis	Foundation for detailed design
Reference	Guide throughout project

11.3 Structure of Design Report

1. Executive Summary

2. Introduction
3. System Description
4. System Objectives
5. Constraints
6. Information Analysis
7. Alternative Evaluation
8. Implementation Plan
9. Cost-Benefit Analysis
10. Appendices

11.4 Cost-Benefit Analysis Example

Costs

Cost Category	Amount	Year 1	Year 2	Year 3
Hardware	& 15L	& 15L	-	-
Software	& 10L	& 10L	-	-
Development	& 20L	& 20L	-	-
Training	& 5L	& 5L	-	-
Maintenance	& 5L	& 5L	& 5L	& 5L
Total	& 55L	& 55L	& 5L	& 5L

Benefits

Benefit Category	Annual Benefit
Labor Savings	& 20L
Error Reduction	& 10L
Inventory Savings	& 15L
Improved Decision Making	& 10L
Total Annual Benefit	& 55L

Payback Period: 1 year

12. KEY DIAGRAMS FOR EXAM

Diagram	Description
Planning Levels	Strategic, Tactical, Operational
SWOT Analysis	Strengths, Weaknesses, Opportunities, Threats
Business Strategy-MIS Alignment	Cost Leadership, Differentiation, Focus
MIS Response Framework	Input → Response → Outcome
MIS Planning Process	General → Detailed
Conceptual Design Steps	6-step process flow
Constraint Types	Technical, Economic, Operational, etc.
Information Needs	Operational, Tactical, Strategic
Alternative Solutions	In-house, Off-shelf, Outsourced, Cloud
Design Report Structure	Sections and sub-sections

13. IMPORTANT DEFINITIONS

Term	Definition
Business Planning	Process of defining objectives and strategies
Strategic Planning	Long-term organizational direction
SWOT Analysis	Strengths, Weaknesses, Opportunities, Threats
MIS Response	How MIS adapts to business needs
MIS Planning	Planning information systems
Conceptual Design	High-level system definition
System Objectives	What the system should achieve
System Constraints	Limitations on the system
Information Needs	What information is required
Design Report	Documentation of conceptual design
Feasibility Analysis	Assessment of viability
SMART Objectives	Specific, Measurable, Achievable, Relevant, Time-bound

14. FEASIBILITY STUDY

14.1 Introduction

Definition: A feasibility study is an assessment of the practicality and viability of a proposed system or project. It evaluates whether the system can be successfully implemented given existing resources and constraints.

14.2 Types of Feasibility

Type	Description	Key Questions
Technical Feasibility	Whether existing technology and technical resources are sufficient	Do we have the hardware, software, and technical expertise?
Economic Feasibility	Whether the project is financially viable (cost-benefit analysis)	Will benefits exceed costs? What is the ROI?
Operational Feasibility	Whether the system will work within the organization's operations	Will users accept the system? Will it fit current processes?
Schedule Feasibility	Whether the project can be completed within the given timeline	Can we meet the deadline? Are milestones realistic?
Behavioral Feasibility	Whether users and stakeholders will accept and support the system	Will there be resistance to change? Is training adequate?

14.3 Importance of Feasibility Study

- Reduces risk of project failure
- Provides basis for go/no-go decision
- Identifies potential problems early
- Helps in resource allocation
- Ensures alignment with business goals

15. PREVIOUS YEAR QUESTIONS

Long Questions (10 marks)

- Explain business planning and its relationship with MIS planning.
- What is SWOT analysis? How is it used in MIS planning?
- Discuss the process of MIS planning - general and detailed.
- Explain the conceptual system design process with all steps.
- What are system objectives and constraints? Explain with examples.
- Describe how to determine information needs and sources for a system.
- Explain the process of developing alternative conceptual designs and selecting the best one.
- What is a design report? Explain its structure and contents.

Short Questions (5 marks)

- What is the difference between general and detailed MIS planning?
- Explain SWOT analysis in the context of MIS.
- What are system objectives? How are they set?
- What are system constraints? Give examples.
- How do you determine information needs?
- What are the different types of alternative solutions?
- What is the purpose of a design report?
- Explain the SMART criteria for objectives.

What is the difference between internal and external information sources?

What is a cost-benefit analysis in system design?

Very Short Questions (2 marks)

Define business planning.

What is SWOT analysis?

What is an MIS response?

What is conceptual system design?

What is a system objective?

What is a system constraint?

What is a design report?

What is the difference between conceptual and detailed design?

What is a cost-benefit analysis?

What does SMART stand for?

16. SUMMARY OF UNIT III

Topic	Key Points
Business Planning	Strategic, Tactical, Operational levels; SWOT analysis
MIS Responses	Strategic, Tactical, Operational responses to business needs
MIS Planning	General (strategic) and Detailed (project) planning
Conceptual Design	6 steps: Problem Definition, Objectives, Constraints, Information Needs, Alternatives, Design Report
Problem Definition	Identify issues, understand causes, define scope
System Objectives	SMART: Specific, Measurable, Achievable, Relevant, Time-bound
System Constraints	Technical, Economic, Operational, Schedule, Legal, Organizational
Information Needs	Operational, Tactical, Strategic; Internal and External sources
Alternative Designs	In-house, Off-shelf, Outsourced, Cloud; Evaluation matrix
Design Report	Comprehensive documentation with all sections
Feasibility Study	Technical, Economic, Operational, Schedule, Behavioral